Group Architecture and Design

Series: IAMADM | **Notebook:** 3 of 9 | **Created:** January 2026

Designing Scalable Group Structures

Groups are the foundation of access management. A well-designed group architecture simplifies administration, improves security, and scales with your organization.

Table of Contents

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM enabled
Permissions	`account-iam-admin` or group management rights
Prior Knowledge	**IAMADM-01** and **IAMADM-02** (SSO configured)

1. Group Fundamentals

Groups in Dynatrace Gen3 IAM are collections of users that share access permissions. Groups exist at the **account level** and can be assigned to one or more environments.

Group Properties

Property	Description	Required
Name	Unique identifier for the group	Yes
Description	Purpose and ownership information	Recommended
Owner	Person responsible for membership	Recommended
Federated ID	SAML group mapping (if SSO)	Optional

What Groups Control

Groups are assigned:
- **Account-level policies** (apply to all environments)
- **Environment-level policies** (scoped to specific environments)
- **Boundaries** (what entities they can see)

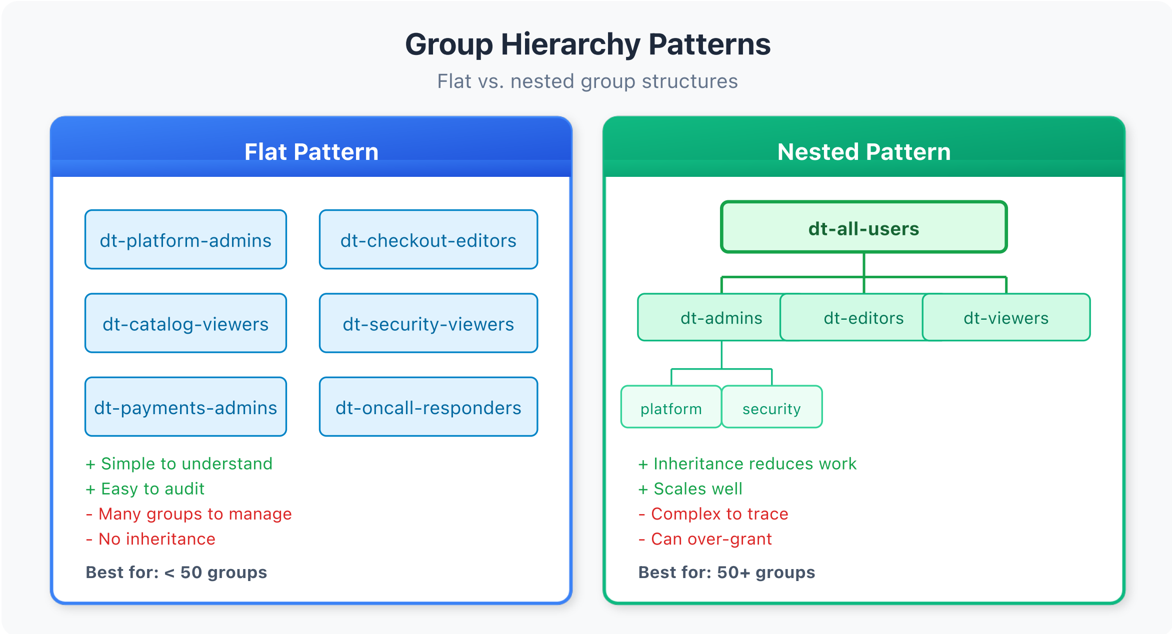
Group vs User Assignment

Approach	Use When	Maintenance
Group-based	Standard case for all users	Low - manage membership only
Direct user	Break-glass, temporary access	High - track individual grants

> **Best Practice:** Always prefer group-based access. Direct user assignment should be rare and time-limited.

2. Group Hierarchy Patterns

Choose a structure that matches your organization and scales appropriately.



Pattern 1: Flat Structure

All groups exist at the same level with no hierarchy.

```
Groups:
├── platform-admins
├── platform-viewers
├── team-a-users
├── team-b-users
└── auditors
```

Pros: Simple, easy to understand
Cons: Doesn't scale, no inheritance

Pattern 2: Role-Based

Groups organized by function/role.

```
Groups:
├── dt-admins              (full access)
├── dt-editors             (read + write)
├── dt-viewers             (read only)
└── dt-auditors            (compliance access)
```

Pros: Clear permission levels
Cons: Limited data segregation

Pattern 3: Team-Based

Groups organized by team ownership.

```
Groups:
├── checkout-team          (checkout service owners)
├── payments-team          (payments service owners)
├── platform-team          (infrastructure owners)
└── sre-oncall             (incident response)
```

Pros: Maps to ownership model
Cons: Permission levels mixed within groups

Pattern 4: Matrix (Recommended for Enterprise)

Combines role and team dimensions.

```
Groups:
├── checkout-admins        (checkout team, admin access)
├── checkout-editors       (checkout team, edit access)
├── checkout-viewers       (checkout team, view access)
├── payments-admins
├── payments-editors
├── payments-viewers
├── platform-admins        (cross-cutting admin)
└── all-viewers            (everyone, read-only)
```

Pros: Fine-grained control, scales well
Cons: More groups to manage

Choosing a Pattern

Organization Size	Recommended Pattern
< 50 users	Flat or Role-Based
50-500 users	Team-Based or Matrix
> 500 users	Matrix

3. Naming Conventions

Consistent naming makes groups discoverable and self-documenting.

Recommended Format

```
&lt;scope&gt;-&lt;team/domain&gt;-&lt;role&gt;
```

Examples:

Group Name	Scope	Team	Role
`dt-checkout-admins`	Dynatrace	Checkout	Admin
`dt-payments-viewers`	Dynatrace	Payments	Viewer
`dt-platform-editors`	Dynatrace	Platform	Editor
`dt-all-oncall`	Dynatrace	All	On-call access

Naming Rules

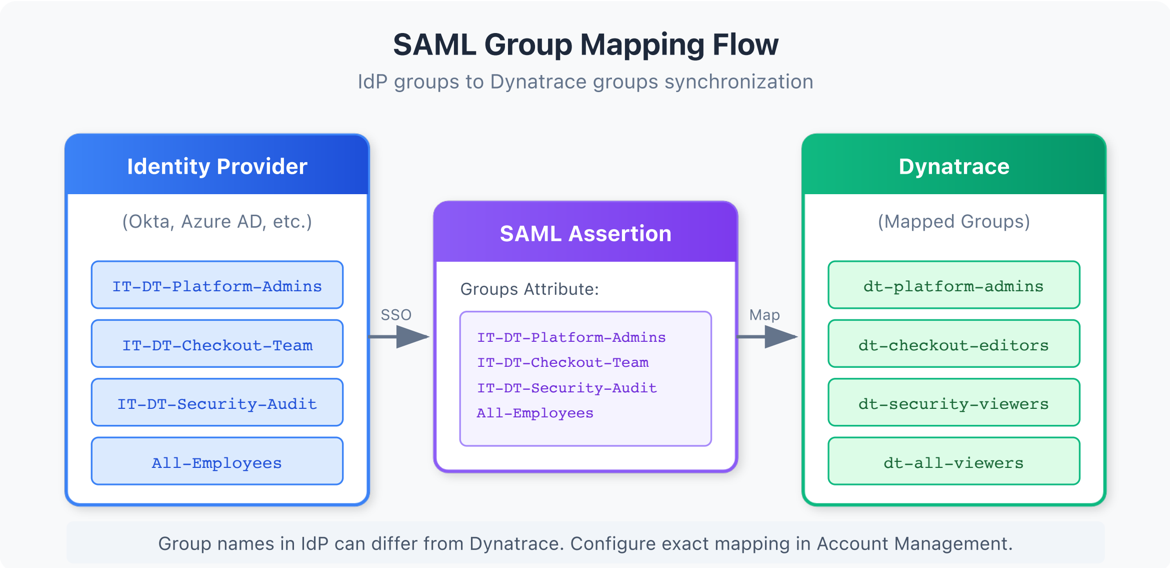
Rule	Rationale
Use lowercase	Consistency, SAML compatibility
Use hyphens (not underscores)	URL-safe, readable
Prefix with `dt-`	Identify Dynatrace groups in IdP
Keep under 50 characters	Display limits
Avoid abbreviations	Clarity over brevity

Anti-Patterns to Avoid

Bad Name	Problem	Better Name
`Team A`	Spaces, generic	`dt-team-a-users`
`ADMINS`	Uppercase, no context	`dt-platform-admins`
`grp_dt_chk_adm`	Cryptic abbreviations	`dt-checkout-admins`
`dynatrace-users`	Too generic	`dt-all-viewers`

4. SAML Group Mapping

Federating groups from your Identity Provider (IdP) automates user provisioning and ensures consistent access.



How It Works

1. User authenticates via SAML SSO
2. IdP includes group memberships in SAML assertion
3. Dynatrace maps IdP groups to Dynatrace groups
4. User receives permissions from mapped groups

Mapping Strategies

Strategy	Description	Use When
1:1 Mapping	IdP group = DT group	Simple, clear ownership
Many:1 Mapping	Multiple IdP groups → one DT group	Consolidating access
1:Many Mapping	One IdP group → multiple DT groups	Granting multiple roles

IdP Group Attribute

Configure your IdP to send groups in the SAML assertion:

Azure AD (Entra ID):

```
Claim name: groups
Source attribute: user.assignedgroups
```

Okta:

```
Attribute: groups
Value: Matches regex: dt-.*
```

OneLogin:

```
SAML Attribute: groups
Value: Member Of (filtered)
```

Mapping Configuration in Dynatrace

In Account Management → Identity providers → Your IdP:

1. **Group claim name**: The attribute name from your IdP (e.g., `groups`)
2. **Group mappings**: Map each IdP group to a Dynatrace group

Example Mappings:

IdP Group	Dynatrace Group
`IT-Dynatrace-Admins`	`dt-platform-admins`
`App-Checkout-Team`	`dt-checkout-editors`
`All-Employees`	`dt-all-viewers`

> **Best Practice:** Use explicit mappings rather than auto-create. This prevents unauthorized groups from being provisioned.

5. Group Lifecycle Management

Groups have a lifecycle that requires ongoing management.

Group Lifecycle Stages

Stage	Actions	Owner
Request	Business justification, approval	Requestor
Create	Create group, configure policies	IAM Admin
Populate	Add initial members	IAM Admin / Group Owner
Operate	Ongoing membership changes	Group Owner
Review	Periodic membership audit	Group Owner + Auditor
Retire	Remove members, delete group	IAM Admin

Group Owner Responsibilities

Every group should have a designated owner who:

- Approves membership requests
- Removes departed employees
- Participates in access reviews
- Maintains group description/documentation

Membership Change Process

Adding Members:
1. User or manager requests access
2. Group owner approves request
3. IAM admin (or automated process) adds user
4. User confirms access works

Removing Members:
1. Triggered by: role change, termination, access review
2. Group owner authorizes removal
3. IAM admin removes user
4. Audit log captures change

Periodic Access Reviews

Group Type	Review Frequency	Reviewer
Admin groups	Monthly	Security + Account Admin
Editor groups	Quarterly	Group Owner + Manager
Viewer groups	Semi-annually	Group Owner

See **IAMADM-07: Audit Logging and Compliance** for access review automation.

6. Separation of Duties

Design your group structure to enforce separation of duties and reduce risk.

Key Separations

Separation	Groups to Separate	Reason
Admin vs User	`dt-platform-admins` vs `dt-all-viewers`	Prevent privilege escalation
Dev vs Prod	`dt-app-dev-*` vs `dt-app-prod-*`	Protect production
Configure vs Operate	`dt-config-editors` vs `dt-operations`	Change control
Audit vs Admin	`dt-auditors` vs `dt-*-admins`	Independent oversight

Incompatible Group Pairs

Users should NOT be in both groups simultaneously:

Group A	Group B	Risk
Production Admins	Development Editors	Dev-to-prod escalation
Security Auditors	Platform Admins	Self-audit
Token Managers	Application Users	Token abuse

Enforcement Options

1. **Manual Review**: Check during access reviews
2. **IdP Rules**: Configure mutually exclusive groups in IdP
3. **Automation**: Script to detect violations

Break-Glass Groups

Emergency access should be handled through special groups:

```
dt-breakglass-admins
├── Empty by default
├── Temporary membership only (&lt; 24h)
├── Requires documented incident
└── Automatic expiration
```

7. Cross-Environment Patterns

When you have multiple Dynatrace environments, design groups that work across them.

Environment Types

Environment	Purpose	Typical Access
Production	Live customer data	Restricted, audit-heavy
Non-Production	Dev, test, staging	Broader access
Sandbox	Learning, experimentation	Open access

Cross-Environment Group Patterns

Pattern A: Environment-Specific Groups

```
dt-prod-checkout-admins    (prod env only)
dt-nonprod-checkout-admins (nonprod env only)
```

Pattern B: Role Groups + Environment Assignment

```
dt-checkout-admins
├── Assigned to Prod env with boundary X
└── Assigned to NonProd env with boundary Y
```

Pattern C: Tiered Groups

```
dt-tier1-admins  (all environments, full access)
dt-tier2-admins  (nonprod only, full access)
dt-tier3-viewers (all environments, view only)
```

Recommendations

Scenario	Recommended Pattern
Strong prod/nonprod separation	Pattern A (env-specific)
Simplified management	Pattern B (role + assignment)
Clear access tiers	Pattern C (tiered)

8. Group Assessment Queries

Use these queries to understand your current group usage.

```
// Review recent group membership changes
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "group") and matchesPhrase(content, "member")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Audit group creation events
fetch logs, from: now() - 90d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "group") and matchesPhrase(content,
"created")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Find group-related changes by admin user
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "group")
| summarize changeCount = count(), by:{content}
| sort changeCount desc
| limit 20
```

Next Steps

With your group architecture defined, proceed to configure access controls:

Recommended Path

1. **IAMADM-04: Policy Authoring and Management** - Create policies for your groups
2. **IAMADM-05: Boundary Design Patterns** - Define what each group can see
3. **IAMADM-06: User Lifecycle and Provisioning** - Automate user management

Group Architecture Checklist

Before moving on, ensure you have:

- [ ] Chosen a group hierarchy pattern
- [ ] Defined naming conventions
- [ ] Configured SAML group mappings (if using SSO)
- [ ] Documented group ownership
- [ ] Planned separation of duties
- [ ] Designed cross-environment patterns (if applicable)

Summary

In this notebook, you learned:

- Group fundamentals and properties
- Four hierarchy patterns: flat, role-based, team-based, and matrix
- Naming conventions for scalable group management
- SAML group mapping strategies
- Group lifecycle management and ownership
- Separation of duties through group design
- Cross-environment group patterns

References

- User Groups
- SAML 2.0 Integration
- Manage User Permissions

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.